

Spectral Tree Inference from Incomplete Matrices: Fiedler Vector Bounds under Sub-sampling

Itay Gonen

Abstract

Top-down reconstruction of latent binary phylogenetic trees is fundamentally bottlenecked by the $\Theta(m^2)$ cost of computing pairwise similarities across m leaves. While observing only a sparse, Bernoulli-sampled subset of entries offers a computational workaround, real-world tree imbalance severely degrades the spectral signal, rendering standard balanced-cluster recovery guarantees inadequate. In this work, we frame spectral tree inference as an eigenvector perturbation problem within the matrix-completion regime, demonstrating that the primary root split of an imbalanced latent tree can be exactly recovered from the sign pattern of the empirical Fiedler vector.

Our primary contribution is a sharp finite-sample guarantee proving that a sampling rate of $p = \Theta(\log m/m)$ suffices for exact split recovery at a fixed imbalance. By utilizing a coordinate-wise Neumann-series expansion rather than a standard Davis-Kahan ℓ_2 transfer, we eliminate a spurious $\sqrt{m}$ factor in the entry-wise error bound. Furthermore, we identify the imbalance ratio η as the governing topological parameter and prove it dictates recovery through two distinct failure mechanisms: artificially inflating subspace coherence (rendering recovery statistically unaffordable) and eroding the spectral gap (rendering recovery structurally impossible). This yields an explicit sample complexity dependence of $(1 + \eta)^3$. Finally, we establish a parallel spectral recovery principle for double-centered distance matrices. Simulations on closed-form block models and Kingman coalescent trees with JC69 sequence evolution validate the predicted $\Theta(\log m/m)$ scaling and the isolated degradation mechanics of topological imbalance.

1 Introduction

A latent tree graphical model relates m observed leaves through unobserved internal nodes, and the inference problem is to recover the topology from leaf data alone [2, 26]. Phylogenetics is the canonical instance and supplies our vocabulary: the leaves are extant organisms, the internal nodes their extinct ancestors, the data aligned sequences. Distance-, likelihood- and spectral-based reconstruction are all well developed [25, 14, 19, 22], but datasets now reach tens of thousands of leaves, and the accurate methods are the slow ones.

That pressure produced the top-down spectral family, which is spectral clustering run on a phylogeny. Build a similarity matrix S over the leaves, form the Laplacian $L = D - S$, split the leaf set by the sign pattern of the Fiedler vector of L , recurse on each side, and merge [2, 30, 16]. The split is provably consistent: the two sides are complementary *clans*, the unrooted analogue of a clade [31], which is what keeps the merge tractable. Both the error and the cost of the recursion land on the top split, since no parent call corrects it and it is the only call that sees all m leaves. Every entry of S is a sequence-pair estimate over a length- ℓ alignment rather than a lookup, so that one call costs $\Theta(m^2)$ estimator evaluations — the cost divide-and-conquer exists to avoid.

This paper asks what happens to the spectrum when that cost is not paid. Compute only a random p -fraction of the pairs, reweight the survivors by $1/p$, and take the Fiedler vector of the

incomplete matrix, in the manner of low-rank matrix completion [9, 10, 8, 24, 20, 6]. Does its sign pattern still recover the split, and how small can p be? There is reason to expect an answer well below 1: under a molecular clock the population matrix has exactly two blocks and the Fiedler vector is constant on each clan, so the signal being sampled is extremely concentrated.

Two things keep this from being routine. What decides recoverability from a random subset is how spread the leading subspace is across coordinates, not how large the matrix is [11]; and guarantees for recursive spectral bipartitioning are stated for clusters of comparable size [4, 23], whereas real phylogenies are markedly more imbalanced than standard null models predict [7, 32, 21, 27, 3]. Imbalance pushes both governing quantities the wrong way: it concentrates the split signal onto the smaller clan, raising the sampling cost, and it shrinks the eigenvalue gap that isolates the Fiedler vector, degrading the signal itself. The two failures are distinct — one leaves the eigenvector informative but unaffordable to sample, the other leaves nothing worth sampling — so pooling them shows one curve where there are two mechanisms.

Recovering a split from a sign pattern is a claim about *every* coordinate of an eigenvector, whereas classical perturbation theory controls only its Euclidean norm [12]; passing between the two charges every node as though it were the worst one, which here decides whether a guarantee is satisfiable at all. We therefore work coordinate-wise [13, 15, 1, 29].

Our contributions are:

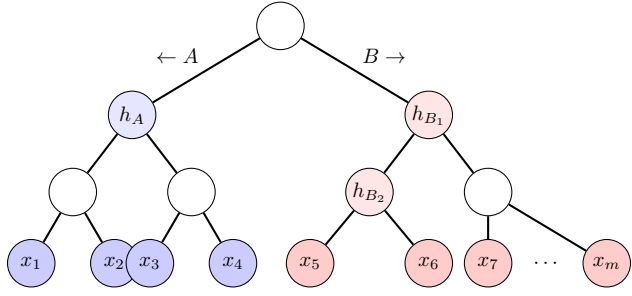
- We make the imbalance ratio η the single parameter through which tree shape reaches the linear algebra, and express both governing spectral quantities in it: the subspace coherence in closed form, and a lower bound on the Fiedler gap (Section 3).
- We prove that a sampling rate of order $\log m/m$ suffices, at fixed imbalance, for the sign pattern of the empirical Fiedler vector to recover the split exactly with high probability, with an explicit cubic dependence on η (Section 4).
- The proof works coordinate-wise rather than in Euclidean norm, through a Neumann expansion and a per-node Bernstein bound (Section 5); on this model the Euclidean route costs a further factor of m in the required rate.
- The same programme on the double-centred *distance* matrix gives a companion statement [18, 5], and we test both in simulation, scoring against the planted split [28, 17] and stratifying by imbalance rather than pooling.

Section 2 fixes the generative model, the observation model and the two operators we analyse; Section 3 routes tree shape into the spectrum through η . Section 4 states the Laplacian rate and ?? the distance rate, Section 5 assembles the proof, and Section 6 tests them.

2 Problem Definition

Our starting point is purely topological. Let $\mathcal{T} = (V, E)$ be a rooted latent binary phylogenetic tree whose vertex set $V = \mathcal{X} \cup \mathcal{H}$ splits into m observed terminal leaves $\mathcal{X} = \{x_1, \dots, x_m\}$ and $m - 1$ unobserved internal ancestral nodes $\mathcal{H} = \{h_1, \dots, h_{m-1}\}$, among them a distinguished root $r \in \mathcal{H}$ [2]. The two edges descending from r define the *primary split* of the tree: the leaf set $\mathcal{X}$ is partitioned into the two subtrees hanging off the root.

We call each side of this split a *clan* [2, 31, 26]: topologically, a clan is the leaf set of one of the two root subtrees, the maximal monophyletic group induced by the root edge. The term is the unrooted analogue of a clade [31], so the root-split bipartition is precisely a pair of complementary



(a) Balanced-Depth Clans $\mathcal{T}$

$$S = \begin{pmatrix} S_A & \beta_0 J_{a \times b} \\ \beta_0 J_{b \times a} & S_B \end{pmatrix}$$

(b) Matrix S Structure

$$L = \begin{pmatrix} D_A - S_A & -\beta_0 J_{a \times b} \\ -\beta_0 J_{b \times a} & D_B - S_B \end{pmatrix}$$

(c) Laplacian L Structure

Figure 1: Visual topology of two lineage clans under a localized root split boundary condition, maintaining the same depth but differing terminal node counts. Shown alongside their corresponding population matrix block configurations (S and L stacked vertically).

clans, consistent with the unrooted similarity matrix we analyze. This single bipartition $\mathcal{X} = A \cup B$, rather than the full branching pattern of $\mathcal{T}$, is the object we set out to recover; resolving it is the top-down step from which the remaining tree structure follows recursively. Throughout, we order the leaves along the split so that leaves 1 through a form clan A and leaves $a + 1$ through m form clan B , which lets us speak of a clan equivalently as a contiguous block of indices.

The bridge from this topology to linear algebra is the population similarity matrix. Let $S \in \mathbb{R}^{m \times m}$ be the symmetric matrix of pairwise similarities over the leaves $\mathcal{X}$, normalized so that $S_{ii} = 1$ [2]. Under the clan ordering, the primary split becomes an algebraic object: the within-clan pairs occupy the two diagonal blocks of S , and the cross-clan pairs its off-diagonal block. Writing $a = |A|$, $b = |B|$, and $J_{a \times b}$ for the all-ones block, the constant cross-clan boundary developed below gives S and its combinatorial Laplacian $L = D - S$ the block form shown in Figure 1. The generative model below makes this matrix concrete, turning branch lengths and the substitution process into the block structure of S .

Branch lengths put a quantitative skin on the bare topology. Each edge $(h, h') \in E$ carries a branch length $\tau_{hh'} \geq 0$ representing evolutionary time. Absolute node times $\tau_h \geq 0$ are measured from the root, with leaves fixed at $\tau = 0$, so $\tau_{hh'} = |\tau_h - \tau_{h'}|$. Running a substitution process along these edges is what converts the tree into the similarity matrix S .

Assumption A1 (Generative model: GTR and log-det similarity). Sequences of length ℓ over a d -character alphabet evolve along $\mathcal{T}$ under a *General Time-Reversible* (GTR) rate matrix $Q \in \mathbb{R}^{d \times d}$ satisfying (i) $Q_{ij} \geq 0$ for $i \neq j$, (ii) $\sum_j Q_{ij} = 0$ for all i , (iii) a stationary distribution $\pi > 0$ with $\pi Q = 0$, and (iv) $\text{diag}(\pi)Q$ symmetric (reversibility). These imply $\text{tr}(Q) < 0$ with all eigenvalues non-positive, and the substitution matrix along (h, h') is $P_{h|h'} = \exp(Q \tau_{hh'})$. The *population similarity* between leaves i, j is the log-det similarity [2]

$$S_{ij} = \exp(4 \text{tr}(Q) \tau_{ij}), \quad (1)$$

where τ_{ij} is the depth of $\text{MRCA}(i, j)$, a purely topological quantity; (1) thus maps the tree onto S , the first fully linear-algebraic object of the paper. The empirical log-det estimator $\hat{S}_{ij}$ concentrates around S_{ij} at rate $\mathcal{O}(1/\sqrt{\ell})$ [2]; the structural conditions that follow are stated as properties of

the population S , whose observable proxy is $\hat{S}$. That choice fixes the scope of the guarantees in Section 4: they carry no $\mathcal{O}(1/\sqrt{\ell})$ term, so the error from a finite alignment is omitted from every rate in this paper, and the sampled entries in (7) are taken to be exact.

We now read the topology off the matrix. Since $\text{tr}(Q) < 0$, (1) makes S_{ij} a decreasing function of the MRCA depth τ_{ij} , so leaves with a shallower common ancestor are more similar. Within-clan pairs satisfy $\tau_{ij} < \tau_{\text{root}}$ and therefore dominate every cross-clan pair: the primary split appears as two strong diagonal blocks of S separated by a weaker off-diagonal block. Two conditions turn this depth monotonicity into algebra: the molecular clock (Assumption A2) collapses the cross-clan block to a single baseline, and the Clan-Floor Block Model (Assumption A3) requires the two diagonal blocks to clear that baseline by a fixed floor, leaving their internal structure free. Between them they also confine every off-diagonal entry to $(0, 1)$, so no separate range restriction is needed (Remark 1). A third condition, the Hierarchical Block Model, derives that floor from tree shape; since no result below depends on it, it is deferred to Section D.

Assumption A2 (Molecular clock). The tree is ultrametric: all cross-clan MRCA depths equal the root depth, so every cross-clan similarity collapses to a common baseline $\beta_0 \in (0, 1)$.

Assumption A3 (Clan-Floor Block Model (CFBM)). The population matrix S satisfies a localized Clan-Floor Block Model at the root split if cross-clan similarities equal the baseline β_0 while intra-clan similarities are bounded away from it from above,

$$S_{ij} = \beta_0 \quad (i \in A, j \in B), \quad \beta_0 < S_{\text{in}}^{\min} \leq S_{ij} \leq S_{\text{max}} < 1 \quad (i \neq j \text{ in the same clan}), \quad (2)$$

the diagonal blocks being otherwise arbitrary [2]. Under a valid CFBM configuration, S and $L = D - S$ accept the block form of Figure 1, with diagonal sub-degrees $D_A = \text{diag}(S_A \mathbf{1}_a) + b\beta_0 I_a$ and $D_B = \text{diag}(S_B \mathbf{1}_b) + a\beta_0 I_b$. We call the special case in which the diagonal blocks are themselves constant the *constant-block* CFBM. The spectral consequences — that $v^{(1)} = \frac{1}{\sqrt{m}} \mathbf{1}_m$ and that the Fiedler vector $v^{(2)}$ is piecewise constant on the split — are *derived* from (2) in Lemma 5 rather than assumed here.

Only the *floor* in (2) is required, not constancy: within-clan MRCA depths vary under the clock for symmetric and asymmetric clans alike, so intra-clan entries are in general non-constant, and the constant-block CFBM is realized only by the degenerate two-star tree of ???. What the floor needs is a lower bound on the deepest within-clan coalescence, and the tree itself supplies one: under geometric per-edge decay the weakest intra-clan similarity is pinned at $S_{\text{in}} \alpha^{D_{\text{max}}}$. That is the Hierarchical Block Model. It is not a competing model and not an extra requirement — it is a *sufficient condition* for (2), it is a hypothesis of no result in this paper, and it is invoked only where the per-edge decay rate α is traced through the spectral gap. We therefore state it, together with the chain of generative assumptions that produces the block structure of S , in Section D.

With S and its block structure in hand, the analysis becomes entirely linear-algebraic, and turns on three scalar parameters. The first is purely topological. With clan sizes $a = |A| \leq b = |B|$ and $a + b = m$, the *imbalance ratio* measures how unevenly the root split divides the leaves,

$$\eta := \frac{b}{a} = \frac{|B|}{|A|} \geq 1, \quad (3)$$

so that $\eta = 1$ is the perfectly balanced split while caterpillar-like lineages reach $\eta = \mathcal{O}(m)$; this parameter plays the role of the balanced-cluster constant of Balakrishnan et al. [4]. The second measures how sharply intra-clan similarities exceed cross-clan ones. Writing $S_{\text{out}}^{\text{max}}$ and $S_{\text{out}}^{\text{min}}$ for

the maximum and minimum similarities across the primary root split, and following the range-restriction framework of [4], we capture the sharpness of the structural partition by the *structural margin*

$$\rho := (S_{\text{in}}^{\min} - S_{\text{out}}^{\max}) - (S_{\text{out}}^{\max} - S_{\text{out}}^{\min}). \quad (4)$$

Under the molecular clock (Assumption A2) cross-clan variance vanishes, $S_{\text{out}}^{\max} = S_{\text{out}}^{\min} = \beta_0$, the penalty term collapses, and the parameter reduces to $\rho = S_{\text{in}}^{\min} - \beta_0$. The third measures how concentrated the leading invariant subspace of L is across nodes. Letting $U = [v^{(1)}, v^{(2)}] \in \mathbb{R}^{m \times 2}$ span the rank-2 invariant subspace of the two leading eigenvalues of L , with $v^{(1)} = \frac{1}{\sqrt{m}} \mathbf{1}_m$ the trivial flat eigenvector and $v^{(2)}$ the structural Fiedler vector, the *subspace coherence* is

$$\mu(U) := \frac{m}{2} \max_{1 \leq i \leq m} \|P_U e_i\|_2^2 = \frac{m}{2} \max_{1 \leq i \leq m} ((v_i^{(1)})^2 + (v_i^{(2)})^2), \quad P_U = UU^\top. \quad (5)$$

The last two are spectral quantities of S and $L = D - S$; Section 3 shows that the first determines both.

The three parameters are not free of one another. The margin and the imbalance trade off directly, and the split is recoverable at all only when the former dominates the latter; we record this as a standing assumption, derived from the spectral-gap bound of Lemma 1 in Section 3.

Assumption A4 (Separation margin / recoverability). Under the CFBM, the primary split is spectrally recoverable only if the structural margin strictly dominates the imbalance-weighted cross-clan boundary,

$$\rho > \eta S_{\text{out}}^{\max}, \quad \text{equivalently} \quad S_{\text{in}}^{\min} > (1 + \eta) S_{\text{out}}^{\max}. \quad (6)$$

This keeps the spectral gap positive ($\Delta\lambda > 0$). Under the relaxation of Assumption A5 the geometric floor on $S_{\text{in}}^{\min}$ plays the same role, with $\alpha^{D_{\max}} > (1 + \eta) S_{\text{out}}^{\max} / S_{\text{in}}$ in place of (6) (Proposition 6).

In practice the population matrix S is hidden, and only a sparse sample of its entries is observed. Following the matrix-completion paradigm [10, 9], we observe entries through a random mask and reweight: let $\Omega \in \{0, 1\}^{m \times m}$ be a symmetric mask whose upper-triangular entries are independent Bernoulli(p), $p \in (0, 1]$, and form the empirical similarity by inverse-probability weighting [8],

$$\hat{S}_{ij} = \begin{cases} \frac{1}{p} \Omega_{ij} S_{ij}, & i \neq j, \\ 1, & i = j. \end{cases} \quad (7)$$

This estimator is unbiased, $\mathbb{E}[\hat{S}] = S$, so $E_L = \hat{L} - L$ is mean zero, where $\hat{L} = \hat{D} - \hat{S}$. The resulting error is missingness rather than additive noise: most pairs are left at zero and the rest rescaled by $1/p$, so it is heavy, structured, and its per-node variance diverges as $p \rightarrow 0$ (Proposition 3).

The Laplacian is not the only operator on these leaves whose spectrum carries the split. A second construction replaces S by the pairwise *distance* matrix $\mathcal{D}$ — distinct from the degree matrix D of $L = D - S$ — built from S by $\mathcal{D}_{ij} = -\log S_{ij}$ (??), and centers it, $\mathcal{B} := H\mathcal{D}H$ with $H = I - \frac{1}{m} \mathbf{1}\mathbf{1}^\top$. There the split sits in the sign pattern of $u^{(1)}$, the eigenvector of $\mathcal{B}$ at the eigenvalue of largest absolute value, rather than in the Fiedler vector of L . ?? works out the population spectrum of $\mathcal{B}$ and shows that the same Bernoulli(p) mask Ω applies to $\mathcal{D}$ unchanged, so both operators are observed under one sampling model and Section 4 states a rate for each.

Problem. Given the sparsely observed estimate $\hat{S}$, recover the primary split (A, B) from the sign pattern of the Fiedler vector $\hat{v}^{(2)}$ of $\hat{L}$. We ask for the minimal sampling rate p at which this top-down partition is correct with high probability. Since $\mathbb{E}[\hat{S}] = S$, the question reduces to an eigenvector-perturbation problem over $E_L = \hat{L} - L$, which the remainder of the paper analyzes.

Recovery is, at its core, an *entry-wise* event. The top-down partition succeeds precisely when the empirical Fiedler vector agrees in sign with its population counterpart at every node, that is, when

$$\left\| \hat{v}^{(2)} - v^{(2)} \right\|_{\infty} < \min_i |v_i^{(2)}|. \quad (8)$$

Condition (8) is the formal recovery event that every bound in the paper is measured against.

3 Topological Scaling and Spectral Gap

The reader now has the population model and the recovery problem. What remains is to explain *why* the topology of the primary split governs recovery. We answer this through the single structural parameter η of (3) and trace its effect on the two spectral quantities the main theorem depends on. Working with the clean population Laplacian $L = D - S$, both the subspace coherence $\mu(U)$ and the spectral gap $\Delta\lambda = \lambda_3 - \lambda_2$ are fixed by macro-topological features alone, and η moves both against recovery: it first inflates $\mu(U)$, then erodes $\Delta\lambda$.

Proposition 1 (Topological scaling of coherence). *Under Assumption A3, the tree imbalance η maps deterministically to the subspace coherence,*

$$\mu(U) = \frac{1 + \eta}{2}. \quad (9)$$

The identity follows by substituting the Fiedler coordinates of Lemma 5 into (5); the short computation is given in Section A. Thus a balanced split attains the optimal coherence $\mu(U) = 1$, and the signal spreads more unevenly across nodes as η grows; the generated regime of Section 6 confirms the identity on data from the full generative pipeline. The identity is reported as a structural diagnostic: since the noise bound of Lemma 2 is obtained from the row norms of S directly (Section B), $\mu(U)$ does not itself enter the sampling rate, and the η -dependence of (12) arrives through the margin and the gap instead.

The second spectral quantity is the gap $\Delta\lambda = \lambda_3 - \lambda_2$ that isolates the Fiedler vector, and the same imbalance that raises $\mu(U)$ drives it toward zero.

Lemma 1 (Spectral-gap lower bound). *Let L be the combinatorial Laplacian of a similarity matrix satisfying the molecular clock, with structural margin $\rho = S_{\text{in}}^{\min} - S_{\text{out}}^{\max}$. For $m_{\min} = \min(a, b)$ and imbalance η , the spectral gap isolating the Fiedler vector satisfies*

$$\Delta\lambda \geq m_{\min}\rho - \eta m_{\min} S_{\text{out}}^{\max}. \quad (10)$$

The full algebraic-connectivity derivation is given in Section A, building on the eigenpairs of Lemma 5. The bound (10) is purely a CFBM statement, controlling the gap through the abstract minimum intra-clan similarity $S_{\text{in}}^{\min}$ of (2). Under the sufficient condition of Assumption A5 that minimum stops being a free parameter and becomes a function of tree shape: it is pinned to the geometric floor $S_{\text{in}} \alpha^{D_{\max}}$, so depth erodes the gap in the same way imbalance does, and the gap survives only while the decay $\alpha^{D_{\max}}$ still clears the imbalance-weighted boundary — quantified in Proposition 6.

Proposition 1 and Lemma 1 show that the same parameter η acts on both spectral quantities at once, which produces two distinct ways for top-down recovery to fail; together they bound the admissible region of (m, η) . The first is *structural collapse*, $\Delta\lambda \rightarrow 0$, which we state here as a corollary of the gap lemma; the second is *statistical infeasibility*, $p > 1$, which is a consequence of Theorem 1 below. A positive gap requires the structural margin to dominate the imbalance-weighted cross-clan boundary, $m_{\min}\rho > \eta m_{\min} S_{\text{out}}^{\max}$, i.e. $\eta < S_{\text{in}}^{\min} / S_{\text{out}}^{\max}$ — precisely what Assumption A4 records. Translating this ratio into tree distances bounds the imbalance the topology can sustain.

Corollary 1 (Structural imbalance tolerance). *Write similarities as distances, $S_{ij} = e^{-d(x_i, x_j)}$, and let $r(\mathcal{T}) = \max_{i,j} (-\log S_{ij})$ be the diameter and $h(\mathcal{T})$ the depth of $\mathcal{T}$ [2], with $h(\mathcal{T}) < r(\mathcal{T})$. Then a positive spectral gap is possible only if*

$$\eta \leq \mathcal{O}(e^{\mathcal{O}(h(\mathcal{T})) - r(\mathcal{T})}). \quad (11)$$

In particular, for an average Kingman coalescent tree, where $r(\mathcal{T}), h(\mathcal{T}) = \mathcal{O}(\log m)$ [14, 3], this gives $\eta \leq \mathcal{O}(m^c)$ for a model-dependent constant c .

The proof sketch (substituting the diameter/depth bounds into $\eta < S_{\text{in}}^{\min}/S_{\text{out}}^{\max}$) is given in Section A.

So under balanced generative models the gap survives polynomial imbalance. A positive gap is necessary but not sufficient: even when $\Delta\lambda > 0$, the rate demanded by Theorem 1 grows with η , and Corollary 2 locates the imbalance past which that demand exceeds $p = 1$.

4 Main Result

The entry-wise recovery criterion (8) is what the sampling rate has to buy. This section converts it into a sufficient condition on p for the Laplacian $L = D - S$, stated in the structural quantities of Section 3: the margin ρ of (4) and the imbalance η of (3).

Theorem 1 (Exact top-down split recovery). *In a non-balanced CFBM with m terminal nodes and imbalance $\eta \geq 1$, let ρ be the structural margin and let $\hat{v}^{(2)}$ be the empirical Fiedler vector — the unit eigenvector belonging to the second-smallest eigenvalue of the sampled Laplacian $\hat{L} = \hat{D} - \hat{S}$, formed from entries observed with probability p as in the observation model (7). Assume Assumptions A1 to A4. For the top-down partition to succeed (i.e. $\|\hat{v}^{(2)} - v^{(2)}\|_{\infty} < \min_i |v_i^{(2)}|$), it suffices that*

$$p > \frac{8(1 + \eta)^3 \beta_0^2 \log m}{m(\rho - \eta S_{\text{out}}^{\max})^2} = \Theta\left(\frac{\log m}{m}\right), \quad (12)$$

in which case $\text{sign}(\hat{v}^{(2)})$ recovers (A, B) exactly with high probability.

(12) is a *sufficient* condition. No matching lower bound is proved anywhere in this paper, so the $\Theta(\log m/m)$ is the growth order of that sufficient bound and not a claim that the rate is tight or order-optimal.

The distance operator $\mathcal{B} = HDH$ of Section 2 admits a statement of the same shape — entry-wise sign agreement of an empirical structural eigenvector against a piecewise-constant target, at a rate governed by a spectral gap. Since the hypotheses of the two operators are not nested, it is a *sibling* of Theorem 1 rather than a corollary of it; ?? states it as ?? and proves it.

What remains is to record where these sufficient conditions can be met at all.

Corollary 2 (Vacuity of the certificate under imbalance). *Under the hypotheses of Theorem 1, with β_0 and the margin $\rho - \eta S_{\text{out}}^{\max}$ held bounded away from 0 independently of η , the right-hand side of (12) exceeds 1 once $\eta = \Omega(\sqrt[3]{m/\log m})$; in general the threshold is reached when $\eta^3 \beta_0^2 / (\rho - \eta S_{\text{out}}^{\max})^2 = \Omega(m/\log m)$. No admissible sampling rate $p \in (0, 1]$ then satisfies the sufficient condition, so Theorem 1 certifies nothing about the split in that regime.*

5 Proof Outline

The proof of Theorem 1 compares two quantities evaluated at the same node: the entry-wise error $|\hat{v}_i^{(2)} - v_i^{(2)}|$ and the entry-wise margin $|v_i^{(2)}|$. It runs in three stages — population geometry fixes the margin and noise budget, a resolvent expansion reduces the error at each node to a scalar, and a per-node Bernstein bound controls that scalar — after which imposing (8) yields the rate (12). Each stage is stated as a lemma; the proofs are in Sections A to C.

Stage 1 (population geometry and noise budget). The same imbalance η sets both sides of (8) at once — it fixes the margin and the spectral gap through the piecewise-constant Fiedler vector of Lemma 5 and the gap bound of Lemma 1, kept positive by Assumption A4, while on the other side matrix Bernstein applied to the sampling mask bounds the noise the gap has to absorb.

Lemma 2 (Laplacian spectral error via matrix Bernstein). *Under Assumptions A2 and A3 and the observation model (7), there are constants $\tilde{C}, C_0$ such that if $p \geq C_0 \frac{\log m}{m}$ then with high probability*

$$\|E_L\|_2 \leq \tilde{C} \sqrt{\frac{m \log m}{p}}.$$

The proof is in Section B. Compared against the gap bound of Lemma 1, this is what certifies the strong-signal regime $\|E_L\|_2 < \Delta\lambda$ that the next stage assumes. Under imbalance the margin and the gap are extremal at different nodes, which is what makes the final synthesis η -dependent.

Stage 2 (entry-wise reduction). Since (8) is an ℓ_∞ event while Davis–Kahan controls only ℓ_2 , and the sole dimension-free transfer between them, $\|\cdot\|_\infty \leq \|\cdot\|_2$, charges every node as though it were the worst one, the reduction has to be coordinate-wise: in the regime of Stage 1 a resolvent (Neumann) expansion of $\hat{v}^{(2)}$ in E_L trades the operator norm for a single scalar per coordinate, at the price of a remainder that is second order in the noise-to-gap ratio.

Lemma 3 (Neumann decomposition of the Fiedler vector). *Let L be symmetric with eigenpairs $\{(\lambda_\ell, v^{(\ell)})\}$, let $\hat{L} = L + E_L$ be a symmetric perturbation, and write $\hat{v}^{(2)} = v^{(2)} + w$. If $\|E_L\|_2 < \Delta\lambda$, then, with Π the spectral projector onto the complement of $v^{(2)}$, w admits the exact resolvent expansion $w = \sum_{k \geq 1} (-1)^k (\Delta\lambda^{-1} \Pi E_L)^k v^{(2)}$, whose first-order term is*

$$w^{(1)} = \sum_{\ell \neq 2} \frac{-\langle v^{(\ell)}, E_L v^{(2)} \rangle}{\lambda_\ell - \hat{\lambda}_2} v^{(\ell)}, \quad (13)$$

so that entrywise $w_i = -(E_L v^{(2)})_i / \Delta\lambda + R_i$ with the remainder of second order, $\|R\|_\infty = O(\|E_L\|_2^2 / \Delta\lambda^2)$.

The expansion is due to Eldridge et al. [13], Fan et al. [15] and Abbe et al. [1] rather than to this paper, which is why it is stated as a lemma; the decomposition and its first-order bound are proved in Section C.1, and the remainder R separately in Section C.3 (Lemma 9), its discard being what confines the argument to the strong-signal regime of Stage 1.

Stage 3 (per-node concentration). The surviving scalar $(E_L v^{(2)})_i$ is a sum of independent, mean-zero increments, each carrying the factor $v_i^{(2)} - v_j^{(2)}$. Every within-clan increment therefore vanishes *exactly* — this is a cancellation, not a bound, and it rests on the clock (Assumption A2) making $v^{(2)}$ exactly piecewise constant, so it is the one step that would degrade if the cross-clan block were only approximately constant. Only cross-clan pairs survive, Bernstein’s inequality applies node by node, and since this is the only place the sampling rate p enters the final condition, we state the bound in the main text.

Lemma 4 (Per-node concentration). *With scale $R_B = \beta_0(1 + \eta)/(p\sqrt{m\eta})$, for each $i \in B$ and $\delta > 0$, with probability $\geq 1 - \delta$, $|(E_L v^{(2)})_i| \leq 2\sqrt{2\sigma_B^2 \ln(2/\delta)} + \frac{2}{3}R_B \ln(2/\delta)$. In the variance-dominated regime ($p = \Theta(\log m/m)$), a union bound over all m nodes at $\delta = \delta_0/m$ gives, with high probability uniformly over i ,*

$$|(E_L v^{(2)})_i| \lesssim \sqrt{\frac{8(1 + \eta)\beta_0^2 \log(2m/\delta_0)}{\eta p}}. \quad (14)$$

The increment representation, the cancellation of the within-clan pairs, the per-node variance σ_B^2 and the proof are in Section C.2. This is the step that realises the $\sqrt{m}$ saving that the Davis–Kahan route forfeits: the variance of the increment sum is smaller than $\|E_L\|_2^2$ by a factor m , and it is this per-coordinate saving that turns (8) into the rate (12); Section E runs the two routes side by side and Remark 3 reports the ratio. The three stages now combine.

Proof of Theorem 1. Step 1 (population geometry). Lemmas 1 and 5 supply the right-hand side of (8) as $\min_i |v_i^{(2)}| = c_B$ and a spectral gap $\Delta\lambda$ that is positive under Assumption A4, and Lemma 2 places $\|E_L\|_2$ below it.

Step 2 (entry-wise reduction). Lemma 3 is therefore applicable and gives $w_i = -(E_L v^{(2)})_i/\Delta\lambda + R_i$; discarding R as lower order against c_B reduces (8) to the per-node condition $|(E_L v^{(2)})_i| < \Delta\lambda |v_i^{(2)}|$ holding at every node i .

Step 3 (concentration). Lemma 4 bounds the left-hand side of that condition uniformly over the m nodes.

Step 4 (synthesis). The per-node condition is tightest on clan B , since $\eta \geq 1$. Squaring $\sqrt{8\sigma_B^2 \log m} < \Delta\lambda c_B$ and substituting σ_B^2 , $c_B^2 = 1/(m\eta)$ and the gap bound $\Delta\lambda \geq m_{\min}(\rho - \eta S_{\text{out}}^{\max})$, the factor η cancels and solving for p gives exactly (12) (Proposition 4 and Corollary 4). Under (12) every node satisfies $|w_i| < |v_i^{(2)}|$, hence $\hat{v}_i^{(2)}$ and $v_i^{(2)}$ share a sign; since $v^{(2)}$ is sign-constant on each clan (Lemma 5), the zero-threshold partition of $\hat{v}^{(2)}$ is (A, B) exactly. $\square$

?? states the counterpart of Theorem 1 for the centered distance operator. Its proof follows the same four steps, with the population geometry of Step 1 and the variance load of Step 3 replaced by their distance analogues; Lemma 3 is applied unchanged, being a statement about an arbitrary symmetric perturbation of an arbitrary symmetric matrix. ?? carries the details.

6 Empirical Results

We test both sampling rates in two complementary regimes. In the *synthesized* regime the similarity matrix S is built directly from the closed-form constant-block CFBM (Assumption A3), isolating the linear algebra with no estimation noise. In the *generated* regime dendropy simulates Kingman coalescent trees and evolves JC69 sequences of length $\ell = 10^4$, from which S is the log-det similarity (Assumption A1); this probes whether the two hypotheses the proof actually rests on — the exact cross-clan baseline of Assumption A2 and the separation margin of Assumption A4 — survive realistic tree shapes and finite-sequence noise. The intra-clan structure that the constant-block CFBM idealizes away is not at issue: (2) leaves it free. Each regime is then run through both operators of Section 2: the Laplacian L of Theorem 1, and the centered distance operator $\mathcal{B} = H\mathcal{D}H$ of ??. Each rounds as its own theorem does — k -means on the normalized Laplacian L_{sym} for the similarity route, the sign pattern of $u^{(1)}$ for the distance route.

Recovery is scored by the Normalized Mutual Information (NMI) of the recovered split $\hat{C}$ against the planted split C^* . NMI is the standard accuracy metric for evaluating spectral-clustering partitions against a known ground truth [17, 23]:

$$\text{NMI}(Y, C) = \frac{2I(Y; C)}{H(Y) + H(C)}, \quad (15)$$

with $I(Y; C)$ the mutual information and $H(\cdot)$ the Shannon entropy. NMI attains its maximum value 1 if and only if the two partitions are identical up to relabeling, and decays as they disagree [28, 17]. Composing this with the sign-recovery guarantee of Theorem 1 fixes the population endpoint of our sweeps.

Corollary 3 (Perfect partition recovery and $\text{NMI} = 1$). *Under the hypotheses of Theorem 1, the spectral partition $\hat{C} = \{\hat{C}_A, \hat{C}_B\}$ obtained by thresholding $\hat{v}^{(2)}$ at zero satisfies $\hat{C} = C^*$ with high probability, and therefore $\text{NMI}(\hat{C}, C^*) = 1$.*

Proof. Theorem 1 guarantees $\|\hat{v}^{(2)} - v^{(2)}\|_\infty < \min_i |v_i^{(2)}|$, so for each node i , $|\hat{v}_i^{(2)} - v_i^{(2)}| < |v_i^{(2)}|$, which forces $\hat{v}_i^{(2)}$ and $v_i^{(2)}$ to share a sign. Under the CFBM the population Fiedler vector is sign-constant on each clan [2] ($v_i^{(2)} > 0$ on A , $v_i^{(2)} < 0$ on B), so the zero-threshold partition of $\hat{v}^{(2)}$ recovers C^* exactly. Exact recovery gives $I(\hat{C}; C^*) = H(\hat{C}) = H(C^*)$, whence by (15) $\text{NMI}(\hat{C}, C^*) = 1$. $\square$

All four figures below share one layout: one pair of panels per imbalance, at $\eta = 1, 5, 10, 15$ throughout. On the left, measured NMI against the sampling rate p , one curve per size, the curves running dark to light (viridis) with increasing size, and the red dashed line at the 0.90 level that defines the empirical threshold $\hat{p}^*$. On the right, $\hat{p}^*$ against size on log-log axes, with a dotted reference of the shape the governing rate predicts — $\log m/m$, from (12) on the similarity side and (??) on the distance side — fitted to the measured points. Filled markers are measured; open markers are bounds rather than thresholds, a read-off pinned at either end of the p grid, and are excluded from the fit.

Along the similarity route the transition sharpens and shifts left as the size grows, and $\hat{p}^*$ decays parallel to the reference at every η in both regimes: with no estimation noise in Figure 2, and on Kingman trees in Figure 3, where the coherence and gap identities of Proposition 1 and Lemma 1 continue to hold despite off-model tree shapes and JC69 noise (Assumption A1). The $\Theta(\log n/n)$ sample complexity predicted by (12) therefore survives realistic finite-sequence estimation noise.

Run through $\mathcal{B}$ on the same data, the sign rule of ?? reproduces that decay at $\eta = 1$ and degrades progressively away from balance, until at $\eta = 15$ most synthesized sizes never recover below full data (Figures 4 and 5). Only the $\eta = 1$ panel of Figure 4 lies inside the configuration where hypotheses (i)–(ii) of ?? are verified (??); the remaining panels measure the estimator rather than certify the rate. Which operator tolerates more noise is not read off these sweeps — the two routes differ in rounding rule as well as in operator — and ?? compares the two rates’ parameter groups, not the two estimators.

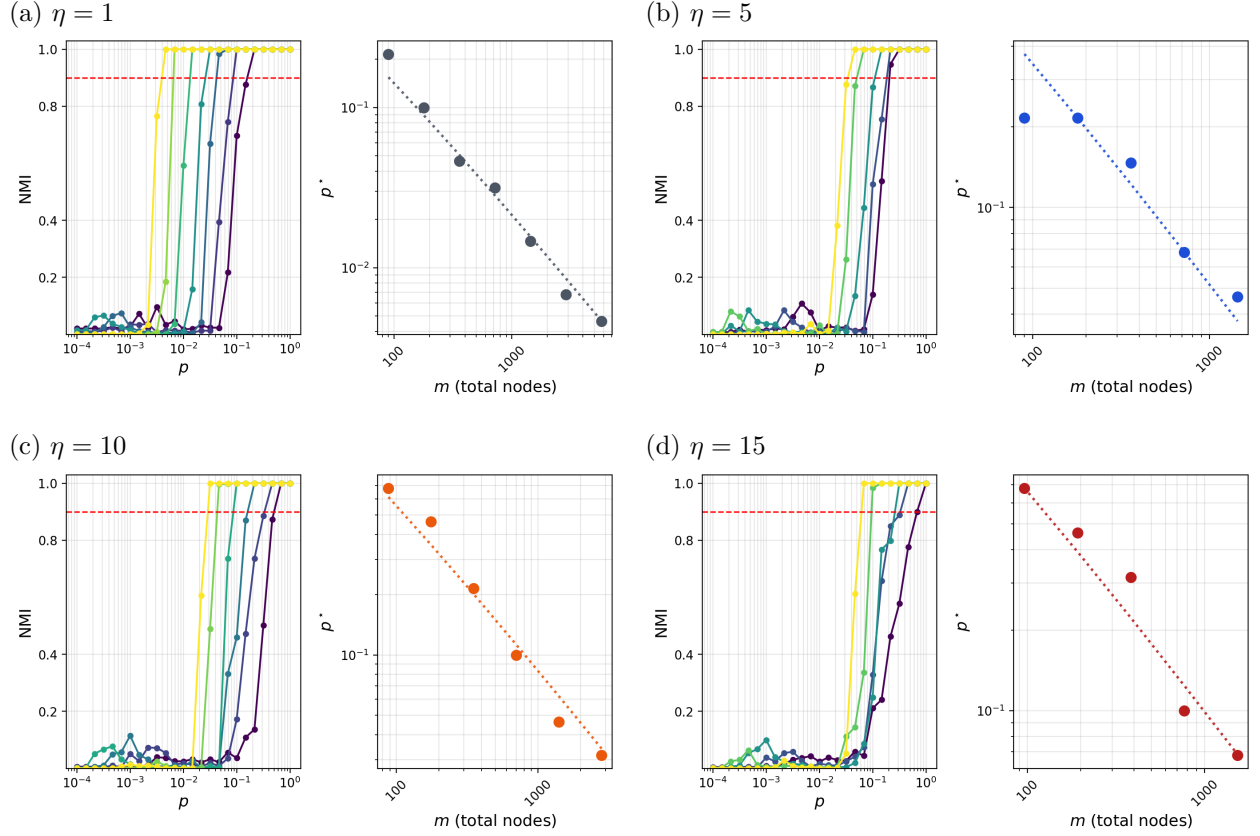


Figure 2: **Similarity route, synthesized non-balanced constant-block CFBM.** $S_{\text{in}} = 0.9$, $S_{\text{out}}^{\text{max}} = 0.05$; seven sizes across two decades of m ; 20 sub-sampling trials per point; k -means rounding. No estimation noise is present: S is the closed-form block matrix.

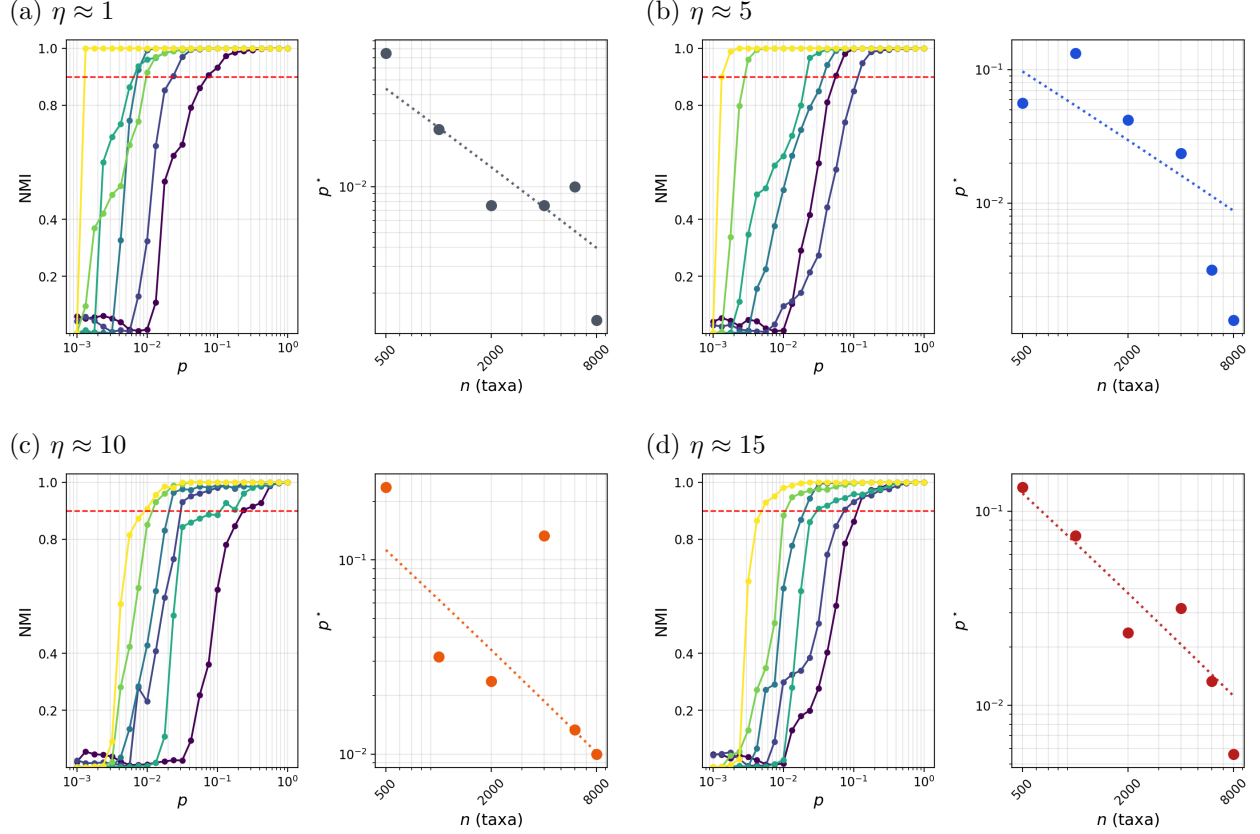


Figure 3: **Similarity route, generated Kingman trees.** JC69 sequences of length $\ell = 10^4$, log-det similarity; $n = 500, 1000, 2000, 4000, 6000, 8000$; k -means rounding. Each point averages the pool samples available at that (n, η) — ten at $\eta \approx 1, 5$, three at $\eta \approx 10, 15$, and one at $n = 8000$ throughout.

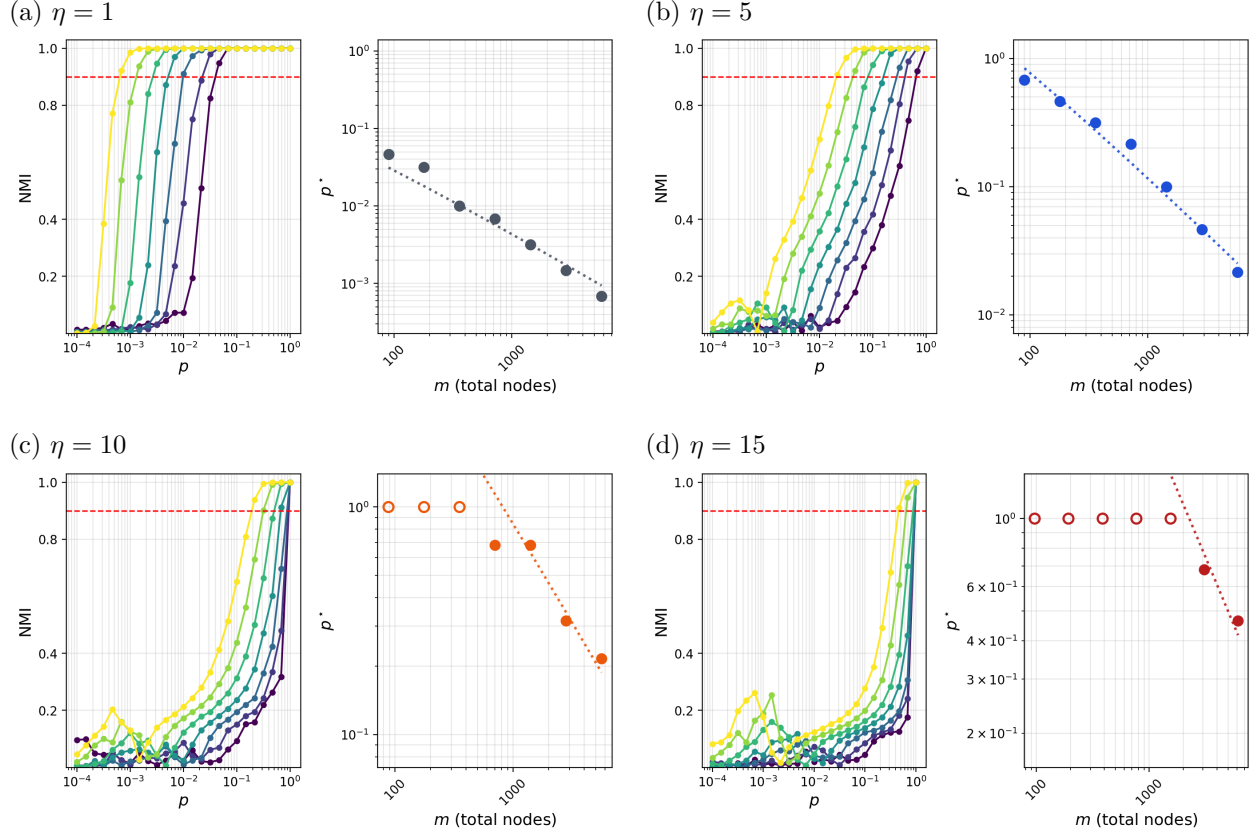


Figure 4: **Distance route, synthesized non-balanced constant-block CFBM.** The matrix of Figure 2 read through $\mathcal{B} = HDH$ with $\mathcal{D} = -\log S$; sign rounding; 8 trials per point.

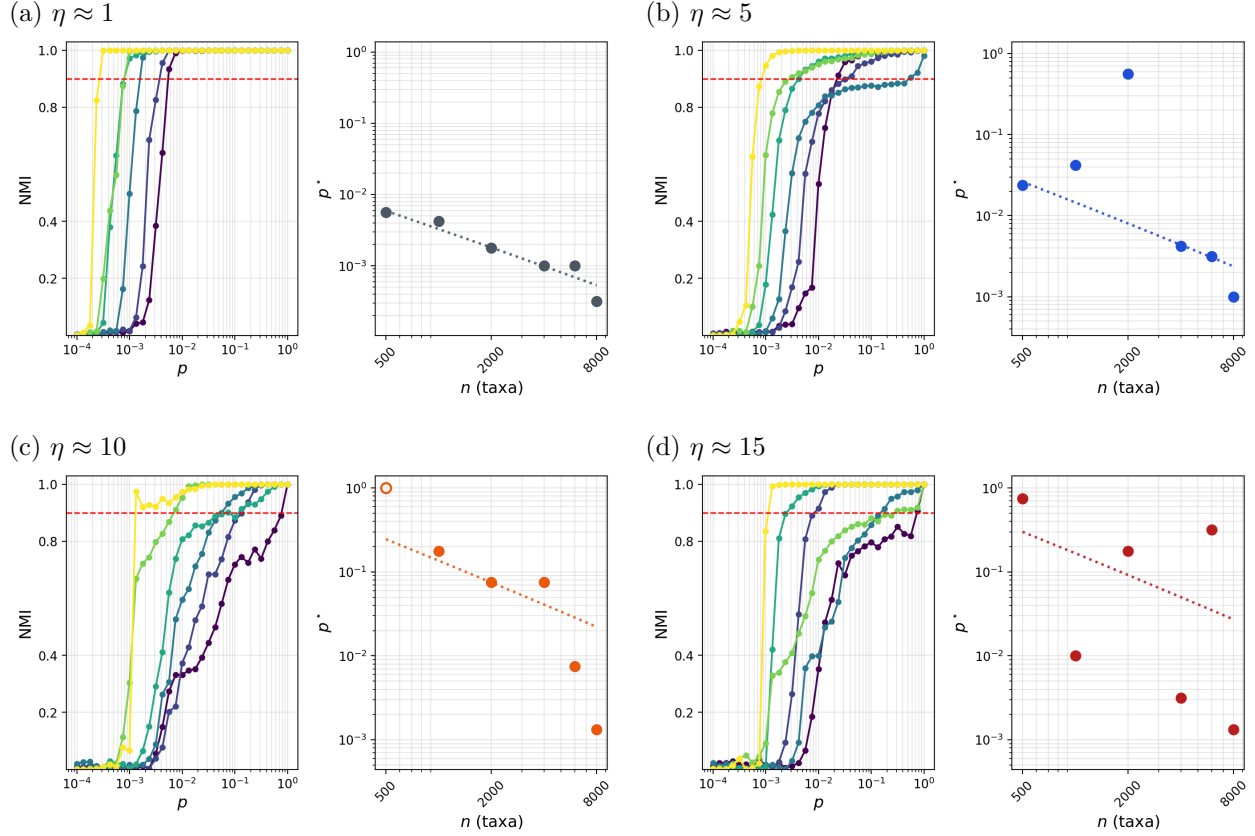


Figure 5: **Distance route, generated Kingman trees.** The same η -binned pool and the same p grid as Figure 3 — same trees, same sequences, same ten sub-sampling replicates per point — read through $\mathcal{B}$ with $\mathcal{D} = -\log S$ recovered from each cached S ; sign rounding. The replicates are combined as in Figure 3: sign-align, average the eigenvectors, partition the average *once*.

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A Proof of the spectral geometry

We first record the range of the population entries, which earlier versions carried as a standing assumption and which is in fact a consequence of the three that precede it.

Remark 1 (Range of the population entries). Under Assumptions A1 to A3 the population matrix satisfies

$$S_{ii} = 1, \quad 0 < \beta_0 \leq S_{ij} \leq S_{\max} < 1 \quad (i \neq j), \quad S_{\max} := \max_{i \neq j} S_{ij}.$$

Indeed $\tau_{ii} = 0$ makes $S_{ii} = \exp(0) = 1$ — consistent with (7), whose unbiasedness $\mathbb{E}[\hat{S}] = S$ requires it — while off the diagonal Assumption A2 places every cross-clan entry at $\beta_0 \in (0, 1)$ and (2) places every same-clan entry in $(\beta_0, S_{\max}]$. No range claim is made for $\hat{S}$: by (7) its observed off-diagonal entries equal S_{ij}/p , which exceed 1 throughout the regime $p = \Theta(\log m/m)$ of interest.

Every later appeal to a bounded entry is to this remark. For the proof of Lemma 1 we next derive the spectral properties of the Fiedler vector.

Lemma 5 (Spectral decomposition of the imbalanced Laplacian). *Under Assumptions A2 and A3:*

(i) *the smallest eigenvalue is $\lambda_1 = 0$ with $v^{(1)} = \frac{1}{\sqrt{m}}\mathbf{1}_m$; (ii) the Fiedler eigenvalue is $\lambda_2 = m\beta_0$; (iii) the Fiedler vector is piecewise constant, $v_i^{(2)} = +c_A$ for $i \in A$ and $v_i^{(2)} = -c_B$ for $i \in B$, with $c_A = \sqrt{b/(am)}$ and $c_B = \sqrt{a/(bm)}$.*

Proof. For any $L = D - S$, $L\mathbf{1} = D\mathbf{1} - S\mathbf{1} = 0$, so $\lambda_1 = 0$ with $v^{(1)} = \frac{1}{\sqrt{m}}\mathbf{1}_m$; no assumption on S is needed.

Step heights. Assume $v^{(2)}$ has the stated form with $c_A, c_B > 0$. The constraints $\|v^{(2)}\|_2 = 1$ and $\langle v^{(1)}, v^{(2)} \rangle = 0$ give $ac_A = bc_B$, hence

$$a\left(\frac{b}{a}c_B\right)^2 + bc_B^2 = 1 \implies \frac{bm}{a}c_B^2 = 1 \implies c_B^2 = \frac{a}{bm}, \quad c_A^2 = \frac{b}{am}.$$

Eigenpair verification. Write $v^{(2)} = (c_A\mathbf{1}_a, -c_B\mathbf{1}_b)^\top$ with $ac_A = bc_B$. The only structural input is that the diagonal sub-degrees of Assumption A3 already carry the row sums of their own block, $D_A = \text{diag}(S_A\mathbf{1}_a) + b\beta_0\mathbf{I}_a$, so that

$$(D_A - S_A)\mathbf{1}_a = b\beta_0\mathbf{1}_a, \quad (D_B - S_B)\mathbf{1}_b = a\beta_0\mathbf{1}_b, \tag{16}$$

identically, for arbitrary symmetric S_A, S_B : no constancy of the diagonal blocks is used, and the intra-clan floor of (2) plays no role here. Using the block form of L with $J_{a \times b}\mathbf{1}_b = b\mathbf{1}_a$, $J_{b \times a}\mathbf{1}_a = a\mathbf{1}_b$ and (16),

$$Lv^{(2)} = \begin{pmatrix} b\beta_0c_A\mathbf{1}_a + b\beta_0c_B\mathbf{1}_a \\ -a\beta_0c_A\mathbf{1}_b - a\beta_0c_B\mathbf{1}_b \end{pmatrix},$$

which reduces to $(b\beta_0(c_A + c_B)\mathbf{1}_a, -a\beta_0(c_A + c_B)\mathbf{1}_b)^\top$. From $ac_A = bc_B$, $b(c_A + c_B) = mc_A$ and $a(c_A + c_B) = mc_B$, so $Lv^{(2)} = m\beta_0 v^{(2)}$ and $\lambda_2 = m\beta_0$. $\square$

Proof of Proposition 1. Substituting the Fiedler coordinates of Lemma 5 into (5),

$$\mu(U) = \frac{m}{2} \max\left\{\frac{1}{m} + c_A^2, \frac{1}{m} + c_B^2\right\} = \frac{m}{2} \left(\frac{1}{m} + \frac{b}{ma}\right) = \frac{1}{2} \left(1 + \frac{b}{a}\right) = \frac{1+\eta}{2}. \quad \square$$

Proof of Lemma 1. The argument uses Assumptions A2 and A3 and Remark 1. By Lemma 5 the Fiedler eigenvalue is $\lambda_2 = m\beta_0 = mS_{\text{out}}^{\max}$. For λ_3 , standard algebraic-connectivity interlacing gives $\lambda_3(L) \geq \min(\lambda_2(L_{C_1}), \lambda_2(L_{C_2}))$, and applying the Fiedler-eigenvalue formula to each sub-clan Laplacian gives $\lambda_2(L_{C_c}) \geq n_c S_{\text{in}}^{(c)}$. With $S_{\text{in}} = \min_c S_{\text{in}}^{(c)}$,

$$\min(\lambda_2(L_{C_1}), \lambda_2(L_{C_2})) \geq \min(n_1, n_2) S_{\text{in}} = m_{\min} S_{\text{in}},$$

so $\Delta\lambda = \lambda_3 - \lambda_2 \geq m_{\min} S_{\text{in}} - m S_{\text{out}}^{\max}$. Using $S_{\text{in}} = \rho + S_{\text{out}}^{\max}$ and $m = m_{\min}(1 + \eta)$,

$$\Delta\lambda \geq m_{\min}(\rho + S_{\text{out}}^{\max}) - (1 + \eta)m_{\min} S_{\text{out}}^{\max} = m_{\min}\rho - \eta m_{\min} S_{\text{out}}^{\max}. \quad \square$$

Proof of Corollary 1. Positivity of the bound in Lemma 1 requires $\eta < S_{\text{in}}^{\min}/S_{\text{out}}^{\max}$. The minimum intra-clan similarity is bounded by the diameter, $S_{\text{in}}^{\min} \geq e^{-r(\mathcal{T})}$, and the maximum cross-clan similarity by the subtree depth, $S_{\text{out}}^{\max} \propto e^{-\mathcal{O}(h(\mathcal{T}))}$. Substituting into $\eta < S_{\text{in}}^{\min}/S_{\text{out}}^{\max}$ gives (11); the logarithmic scaling $r, h = \mathcal{O}(\log m)$ under Kingman yields $\eta \leq \mathcal{O}(e^{\mathcal{O}(\log m)}) = \mathcal{O}(m^c)$. $\square$

B Proof of the Laplacian sampling error

We prove Lemma 2 by bounding $\|E_L\|_2 \leq \|E_D\|_2 + \|E_S\|_2$, where $E_S = \hat{S} - S$ and $E_D = \hat{D} - D$. The only structural fact needed is the entry range of Remark 1. By (7) the diagonal is observed exactly, $\hat{S}_{ii} = S_{ii} = 1$, so it carries no noise and only off-diagonal pairs enter the sums below; on those $S_{ij} \leq S_{\max} < 1$, giving

$$\max_i \sum_{j \neq i} S_{ij}^2 \leq (m-1) S_{\max}^2 < m S_{\max}^2. \quad (17)$$

Proof of Lemma 2. Bounding $\|E_S\|_2$. Write $E_S = \sum_{i < j} Z^{(ij)}$, where $Z^{(ij)}$ is supported on the symmetric off-diagonal pair (i, j) with value $(\frac{1}{p}\Omega_{ij} - 1)S_{ij}$; the summands are independent and mean zero. The scale is $\|Z^{(ij)}\|_2 \leq \max(\frac{1}{p} - 1, 1)S_{ij} \leq S_{\max}/p =: R$. For the variance proxy, $\mathbb{E}[(Z_{ij}^{(ij)})^2] = \frac{p(1-p)}{p^2} S_{ij}^2 \leq \frac{1}{p} S_{ij}^2$, so by (17),

$$\sigma^2 \leq \frac{1}{p} \max_i \sum_{j \neq i} S_{ij}^2 \leq \frac{1}{p} m S_{\max}^2.$$

Matrix Bernstein [29], $\mathbb{P}(\|E_S\|_2 \geq t) \leq 2m \exp(\frac{-t^2/2}{\sigma^2 + Rt/3})$, gives in the variance-dominated regime, with probability $1 - \epsilon$, $\|E_S\|_2 \leq 2S_{\max} \sqrt{m \ln(2m/\epsilon)/p}$.

Bounding $\|E_D\|_2$. E_D is diagonal with $(E_D)_{ii} = \sum_{j \neq i} (\frac{1}{p}\Omega_{ij} - 1)S_{ij}$; a scalar Bernstein with $\sigma_D^2 \leq \frac{1}{p} m S_{\max}^2$ and $R_D = S_{\max}/p$ gives $\|E_D\|_2 \lesssim S_{\max} \sqrt{m \ln(2m/\epsilon)/p}$.

Synthesis. The two terms are now of the same order, so

$$\|E_L\|_2 \leq \|E_S\|_2 + \|E_D\|_2 \leq \tilde{C} \sqrt{\frac{m \log m}{p}},$$

with $\tilde{C}$ absorbing $S_{\max}$ and absolute constants. The bound is uniform in η : the imbalance enters the argument only through the gap $\Delta\lambda$ this noise is compared against, not through the noise itself. $\square$

C Proof of the entry-wise perturbation

C.1 Neumann decomposition (proof of Lemma 3)

Write $\hat{L} = L + E_L$ and $\hat{v}^{(2)} = v^{(2)} + w$, and expand w into a resolvent power series in E_L [13].

Lemma 6 (Exact Neumann decomposition). *Define*

$$w^{(1)} = \sum_{\ell \neq 2} \frac{-\langle v^{(\ell)}, E_L v^{(2)} \rangle}{\lambda_\ell - \hat{\lambda}_2} v^{(\ell)}, \quad w^{(2)} = \sum_{\ell \neq 2} \frac{-\langle v^{(\ell)}, E_L w \rangle}{\lambda_\ell - \hat{\lambda}_2} v^{(\ell)} + \langle v^{(2)}, w \rangle v^{(2)}.$$

Then $w = w^{(1)} + w^{(2)}$ exactly, and under $\|E_L\|_2 < \Delta\lambda$ this is the resolvent series $w = \sum_{k \geq 1} (-1)^k (\Delta\lambda^{-1} \Pi E_L)^k v^{(2)}$.

Proof. From $\hat{L}\hat{v}^{(2)} = \hat{\lambda}_2 \hat{v}^{(2)}$ and $Lv^{(2)} = \lambda_2 v^{(2)}$, $(L - \hat{\lambda}_2 I)w = (\hat{\lambda}_2 - \lambda_2)v^{(2)} - E_L v^{(2)} - E_L w$. Projecting onto $v^{(\ell)}$ ($\ell \neq 2$), using $\langle v^{(\ell)}, v^{(2)} \rangle = 0$, gives $\langle v^{(\ell)}, w \rangle = (-\langle v^{(\ell)}, E_L v^{(2)} \rangle - \langle v^{(\ell)}, E_L w \rangle) / (\lambda_\ell - \hat{\lambda}_2)$. Expanding $w = \sum_{\ell} \langle v^{(\ell)}, w \rangle v^{(\ell)}$ and separating the $\ell = 2$ component recovers the decomposition; iterating on $E_L w$ under $\|E_L\|_2 < \Delta\lambda$ yields the convergent series. $\square$

Lemma 7 (Entry-level first-order bound). *Provided $\|E_L\|_2 < \Delta\lambda$, for every node i ,*

$$|w_i^{(1)}| \leq \frac{|(E_L v^{(2)})_i| + \|E_L\|_2 |v_i^{(2)}|}{\Delta\lambda - \|E_L\|_2}. \quad (18)$$

Proof. By Weyl's inequality $|\lambda_\ell - \hat{\lambda}_2| \geq \Delta\lambda - \|E_L\|_2$ for $\ell \neq 2$. Factoring the denominator, $|w_i^{(1)}| \leq |(P_\perp E_L v^{(2)})_i| / (\Delta\lambda - \|E_L\|_2)$ with $P_\perp = I - v^{(2)}(v^{(2)})^\top$. Since $P_\perp x = x - \langle v^{(2)}, x \rangle v^{(2)}$ and $|\langle v^{(2)}, E_L v^{(2)} \rangle| \leq \|E_L\|_2 \|v^{(2)}\|_2^2 = \|E_L\|_2$, $|(P_\perp E_L v^{(2)})_i| \leq |(E_L v^{(2)})_i| + \|E_L\|_2 |v_i^{(2)}|$. $\square$

Remark 2 (Asymmetric correction in the non-balanced CFBM). The correction $\|E_L\|_2 |v_i^{(2)}|$ equals $\|E_L\|_2 c_A$ for $i \in A$ and $\|E_L\|_2 c_B$ for $i \in B$. For the binding clan B this is $\|E_L\|_2 / \sqrt{m\eta}$, strictly sub-dominant to the recovery target c_B when $\|E_L\|_2 \ll \Delta\lambda$. Throughout Section C.2 we therefore use the leading-order simplification $|w_i^{(1)}| \lesssim |(E_L v^{(2)})_i| / \Delta\lambda$ for $i \in B$.

C.2 Per-node Bernstein concentration

Under the leading-order simplification of Remark 2 the entry-wise error at node i is governed by the scalar $(E_L v^{(2)})_i$, which we now represent as a sum of independent increments and concentrate.

Lemma 8 (Laplacian increment representation). *For every node i , $(E_L v^{(2)})_i = \sum_{j \neq i} X_{ij}$ with $X_{ij} = (\frac{\omega_{ij}}{p} - 1)S_{ij}(v_i^{(2)} - v_j^{(2)})$ independent and mean zero.*

Proof. With $(E_S)_{ij} = (\omega_{ij}/p - 1)S_{ij}$, $(E_L)_{ii} = \sum_{k \neq i} (E_S)_{ik}$ and $(E_L)_{ij} = -(E_S)_{ij}$, so $(E_L v^{(2)})_i = \sum_{k \neq i} (E_S)_{ik} v_i^{(2)} - \sum_{j \neq i} (E_S)_{ij} v_j^{(2)} = \sum_{j \neq i} (E_S)_{ij} (v_i^{(2)} - v_j^{(2)})$. $\square$

Proposition 2 (Within-clan terms vanish). *Under Assumption A3, $v^{(2)}$ is constant on each clan (Lemma 5), so $X_{ij} = 0$ for same-clan pairs; only cross-clan pairs survive: $(E_L v^{(2)})_i = \sum_{j \in A} X_{ij}$ for $i \in B$, and $(E_L v^{(2)})_i = \sum_{j \in B} X_{ij}$ for $i \in A$.*

Proposition 3 (Per-node variance). *Under Assumption A3 and the observation model (7), with $\beta_0 = S_{\text{out}}^{\max}$ and using $(c_A + c_B)^2 = m/(ab) = (1 + \eta)^2/(m\eta)$,*

$$\sigma_B^2 := \text{Var}((E_L v^{(2)})_i)_{i \in B} = \frac{(1-p)(1+\eta)\beta_0^2}{\eta p}, \quad \sigma_A^2 = \frac{(1-p)(1+\eta)\beta_0^2}{p} = \eta \sigma_B^2, \quad (19)$$

so clan B has strictly smaller per-node variance; $\sigma_B^2 = (1-p)\tilde{\rho}(\eta)/p$ with $\tilde{\rho}(\eta) = (1+\eta)\beta_0^2/\eta$.

Proof. For $i \in B$ each of the a surviving terms has $S_{ij} = \beta_0$ and $|v_i^{(2)} - v_j^{(2)}| = c_A + c_B$, so $\sigma_B^2 = \frac{1-p}{p}(c_A + c_B)^2 a \beta_0^2$; substituting $a = m/(1+\eta)$ and $(c_A + c_B)^2 = (1+\eta)^2/(m\eta)$ gives the stated value. Clan A follows symmetrically with b in place of a . $\square$

Proof of Lemma 4. By Lemma 8 and proposition 2, for $i \in B$ the quantity $(E_L v^{(2)})_i = \sum_{j \in A} X_{ij}$ is a sum of independent, mean-zero summands with $|X_{ij}| \leq \beta_0(c_A + c_B)/p = R_B$ and total variance σ_B^2 (Proposition 3). Bernstein's inequality for bounded independent summands gives the stated two-term deviation bound at confidence $1 - \delta$. In the variance-dominated regime $p = \Theta(\log m/m)$ the linear term is of lower order, and a union bound over the m nodes at $\delta = \delta_0/m$ yields (14). $\square$

Proposition 4 (Clan B is binding). *Under Assumption A3, $\min_i |v_i^{(2)}| = |c_B|$, attained on the larger clan B. The per-node condition $|(E_L v^{(2)})_i| < \Delta \lambda |v_i^{(2)}|$ requires $p > 8m(1+\eta)\beta_0^2 \log m / \Delta \lambda^2$ on B (the clan-A condition carries an extra $1/\eta$), so since $\eta \geq 1$, clan B binds.*

Proof. Squaring $\sqrt{8\sigma_B^2 \log m} < \Delta \lambda c_B$ and substituting σ_B^2 with $c_B^2 = 1/(m\eta)$: the η cancels, giving $p > 8m(1+\eta)\beta_0^2 \log m / \Delta \lambda^2$. Clan A follows with $c_A^2 = \eta/m$. $\square$

Corollary 4 (Per-node recovery condition). *If $|w_i| < \min_i |v_i^{(2)}|$ for all i then $\text{sign}(\hat{v}^{(2)}) = \text{sign}(v^{(2)})$. Combining Lemma 4 and proposition 4 with the gap bound $\Delta \lambda \geq m(\rho - \eta\beta_0)/(1+\eta)$ (Lemma 1) yields*

$$p > \frac{8(1+\eta)^3 \beta_0^2 \log m}{m(\rho - \eta S_{\text{out}}^{\max})^2} = \Theta\left(\frac{\log m}{m}\right), \quad (20)$$

which is the rate (12) of Theorem 1. This computation is Step 4 of the proof in Section 5.

C.3 The second-order remainder

The bound below is the term discarded at Step 2 of Section 5; it is what restricts the argument to the strong-signal regime $\|E_L\|_2 \ll \Delta \lambda$.

Lemma 9 (Second-order negligibility). $\|w^{(2)}\|_\infty = O(\|E_L\|_2^2 / \Delta \lambda^2)$.

Proof. By Davis–Kahan $\|w\|_2 \leq \|E_L\|_2 / (\Delta \lambda - \|E_L\|_2)$. Each coefficient $b_\ell = -\langle v^{(\ell)}, E_L w \rangle$ satisfies $|b_\ell| \leq \|E_L\|_2 \|w\|_2$, and since $|\lambda_\ell - \hat{\lambda}_2| \geq \Delta \lambda - \|E_L\|_2$ each summand of $w^{(2)}$ contributes $O(\|E_L\|_2^2 / \Delta \lambda^2)$ in ℓ_∞ ; the $v^{(2)}$ component obeys $|\langle v^{(2)}, w \rangle| = O(\|w\|_2^2) = O(\|E_L\|_2^2 / \Delta \lambda^2)$. $\square$

D The Hierarchical Block Model as a sufficient condition

The floor $S_{\text{in}}^{\min}$ of (2) is an abstract property of the matrix. This appendix connects it to the tree: under geometric per-edge decay the floor is pinned by the depth of the deeper clan, which is what makes the gap — and therefore recoverability — a function of tree shape rather than of a free scalar. The hypothesis that does it is the following.

Assumption A5 (Hierarchical Block Model with geometric decay). The HBM instantiates the floor of (2) by letting intra-clan similarities decay geometrically with path depth, parameterized by the maximum tree depth $D_{\max}$, a per-edge decay factor $\alpha \in (0, 1]$ (under GTR with unit edges $\alpha = \exp(4 \operatorname{tr}(Q) \Delta \tau)$), and a top-level baseline S_{in} . We say S satisfies the HBM if the weakest intra-clan similarity is bounded below by $S_{\text{in}}^{\min} \geq S_{\text{in}} \alpha^{D_{\max}}$. The depth $D_{\max}$ is governed by tree shape: under Yule or Kingman models it scales as $\mathcal{O}(\log m)$, while degenerate caterpillar geometries reach $\mathcal{O}(m)$ [27, 7].

Assumption A5 is *added* to Assumptions A1 to A3, none of which is withdrawn. It keeps the constant cross-clan boundary and constrains only the diagonal blocks — exactly the part that (2) leaves free apart from the floor. It is therefore a sufficient condition for Assumption A3 rather than a substitute for it: under the HBM the Fiedler vector is still exactly piecewise constant (Lemma 5), Theorem 1 applies verbatim, and the only quantity that moves is the spectral gap. No result in the main text takes Assumption A5 as a hypothesis. Table 1 places it in the chain that runs from the generative model to the block structure of S ; the rest of the appendix draws the two consequences, a geometric floor on the weakest intra-clan similarity and the spectral-gap bound that floor induces.

Generative assumption	Consequence for S
GTR alone	$S_{ij} = \exp(4 \operatorname{tr}(Q) \tau_{ij})$; diagonal (intra-clan) blocks dominate off-diagonal blocks by MRCA-depth monotonicity.
GTR + molecular clock	Localised CFBM: cross-clan similarities decay uniformly to an exact baseline β_0 .
GTR + clock + intra-clan floor	Full CFBM: within-clan similarities clear the baseline by $S_{\text{in}}^{\min} - \beta_0 > 0$, their internal structure left free; applies to symmetric and asymmetric trees alike.
GTR + clock + path-depth variance	HBM: intra-clan similarities decay geometrically as α^{depth} , which supplies the floor $S_{\text{in}}^{\min} \geq S_{\text{in}} \alpha^{D_{\max}}$ at depth $D_{\max}$.

Table 1: From the generative model to the block structure of S .

Proposition 5 (Geometric floor). *Under Assumption A5, $S_{\text{in}}^{\min} \geq S_{\text{in}} \alpha^{D_{\max}}$ with $D_{\max} = \lceil \log_2 \max(a, b) \rceil$.*

Proof. GTR multiplicativity along paths gives $S_{ij} = \prod_{e \in \text{path}(i,j)} \alpha_e$, so a same-clan pair at depth $\leq D_{\max}$ satisfies $S_{ij} \geq S_{\text{in}} \alpha^{D_{\max}}$; minimising over same-clan pairs gives the floor, with $D_{\max}$ the height of the deeper clan subtree. $\square$

Proposition 6 (HBM spectral-gap bound). *Substituting Proposition 5 into Lemma 1 gives*

$$\Delta \lambda \geq m_{\min} (S_{\text{in}} \alpha^{D_{\max}} - S_{\text{out}}^{\max}) - \eta m_{\min} S_{\text{out}}^{\max} = m_{\min} S_{\text{in}} \alpha^{D_{\max}} - (1 + \eta) m_{\min} S_{\text{out}}^{\max},$$

positive iff $\alpha^{D_{\max}} > (1 + \eta) S_{\text{out}}^{\max} / S_{\text{in}}$. As $\alpha \rightarrow 1$ (or $D_{\max} = 0$) this reduces to Lemma 1 and the CFBM margin condition of Assumption A4.

Figure 6 verifies the bound on simulated HBM matrices, and Figure 7 illustrates the imbalance-driven degradation of the recovered partition.

E The Neumann trick — comparison with standard bounds

The recovery criterion (8) constrains $w = \hat{v}^{(2)} - v^{(2)}$ in the entry-wise sense, $\|w\|_{\infty} < c_B$, where $c_B = \min_i |v_i^{(2)}|$ is the entry-wise margin, attained on the binding clan B with $c_B^2 = 1/(m\eta)$, while

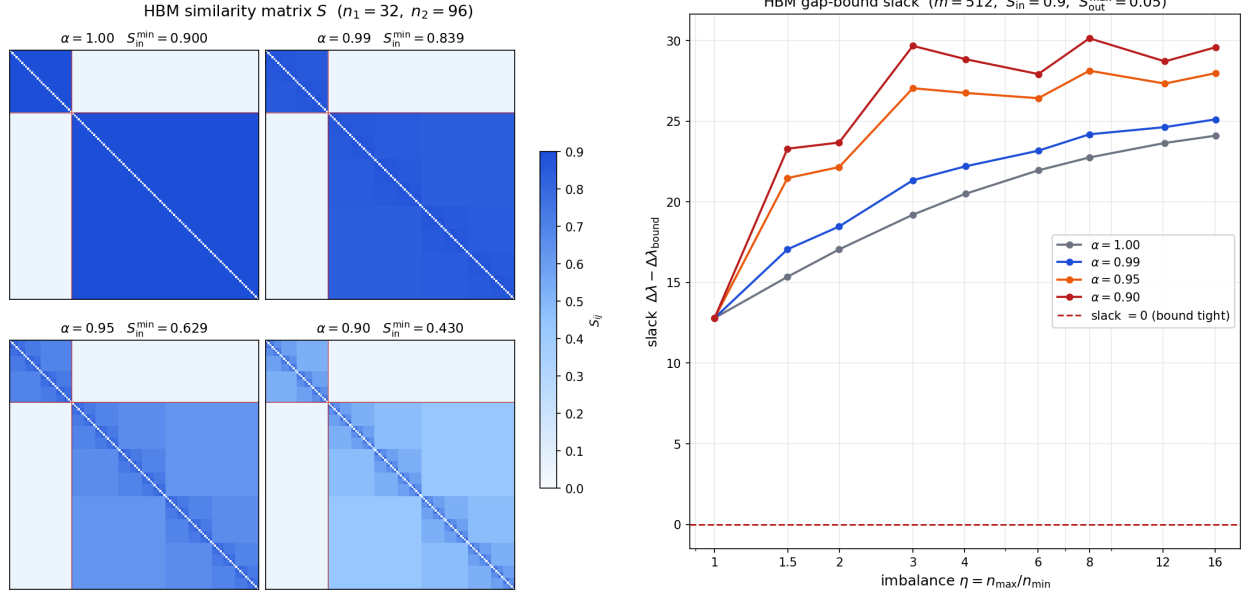


Figure 6: Empirical verification of Proposition 6 over a sweep of (η, m) across decay rates $\alpha \in \{1.0, 0.99, 0.95, 0.90\}$. **(Left)** HBM similarity matrices S degrading as α decreases. **(Right)** slack of the gap lower bound versus η , resolving $S_{in}^{min} = S_{in}\alpha^{D_{max}}$ at $D_{max} = \lceil \log_2 \max(n_1, n_2) \rceil$; the bound stays valid (slack ≥ 0) across all configurations.

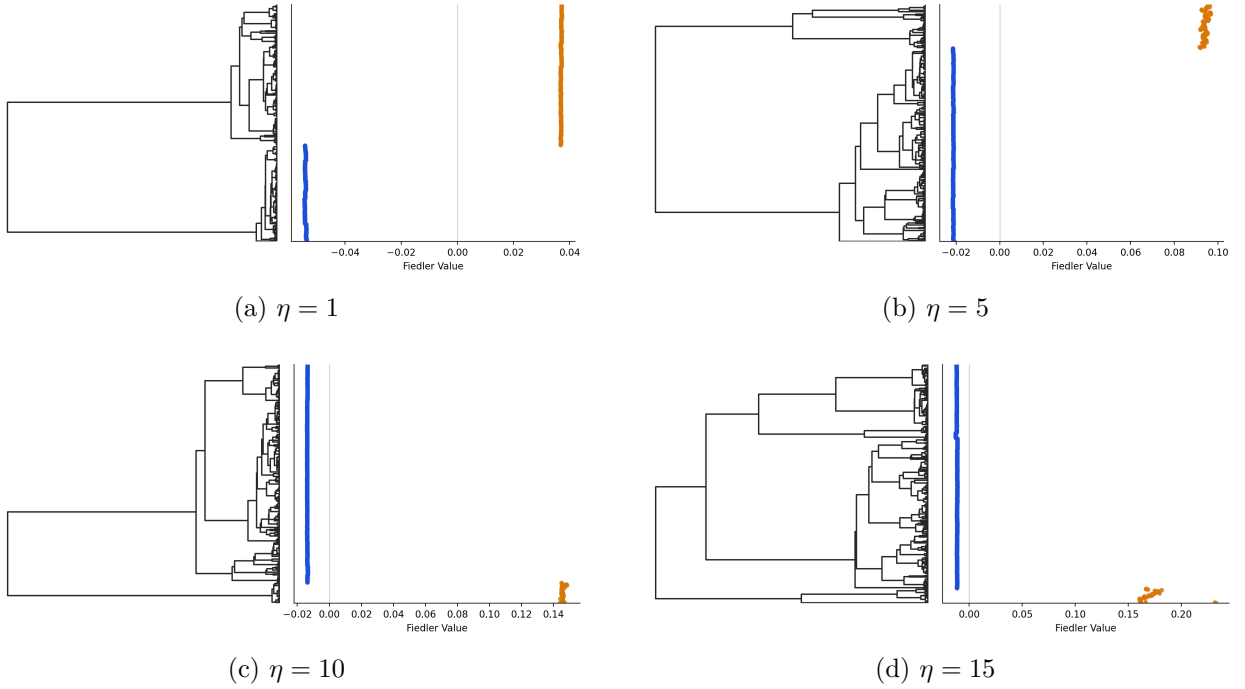


Figure 7: Fiedler bipartitions of a Kingman-model tree with 500 terminal nodes ($n = 10,000, d = 4$, JC69), across increasing imbalance η .

the sampled-Laplacian noise satisfies $\|E_L\|_2 \leq \tilde{C}\sqrt{m \log m/p}$ (Lemma 2). The two routes below bound the same w , differing only in how they relate the entry-wise constraint to E_L .

The standard (ℓ_2) route. Davis–Kahan controls only the Euclidean norm, $\|w\|_2 \leq \|E_L\|_2/(\Delta\lambda - \|E_L\|_2)$. Because the criterion is in the supremum norm, this must be transferred via $\|w\|_\infty \leq \|w\|_2$; imposing $\|w\|_2 < c_B$ and substituting the noise bound in the regime $\|E_L\|_2 \ll \Delta\lambda$ yields $p_{\text{naive}} \gtrsim m^2\eta \log m/\Delta\lambda^2$, i.e. $\eta(1+\eta)^2 \log m/(\rho - \eta\beta_0)^2$ after the gap bound of Lemma 1.

The entry-wise (Neumann) route. Lemma 3 bounds each coordinate directly, replacing $\|E_L\|_2$ by the scalar $(E_L v^{(2)})_i$ at the tightest coordinate $i \in B$. This scalar is a sum of $O(m)$ independent mean-zero contributions concentrating at $\sigma_B^2 = (1-p)(1+\eta)\beta_0^2/(\eta p)$ (Proposition 3), smaller than $\|E_L\|_2^2$ by a factor m ; imposing $|w_i| < c_B$ and the gap bound recovers the rate of Corollary 4.

Remark 3 (ℓ_2 vs. entry-wise). Comparing the two rates gives $p_{\text{naive}}/p_{\text{Neumann}} = \Theta(m\eta/((1+\eta)\beta_0^2))$, a $\Theta(m)$ reduction for fixed η . The saving originates at the noise level: the operator norm $\|E_L\|_2$ exceeds the individual increment $(E_L v^{(2)})_i$ by $\Theta(\sqrt{m})$, and since p enters both rates as p^{-1} under a square root this squares to $\Theta(m)$. Both rates carry $(\rho - \eta\beta_0)^{-2}$ and so diverge together as $\rho - \eta\beta_0 \rightarrow 0$ (the boundary of Assumption A4), but the entry-wise rate stays within $p \leq 1$ over a region where p_{naive} already exceeds it, and under the HBM decay of Proposition 6 remains feasible for larger D_{max} , accommodating deeper and more imbalanced trees than the Euclidean route admits.

F Which Split Each Operator Picks

Two cohorts of 1000 trees, Kingman and birth–death, at $m = 1000$ and $L = 10^4$ under Assumptions A1 and A2, with no sub-sampling. Each operator uses the rounding rule its own theorem states: k -means on the Fiedler vector of $L(S)$, sign on the leading-magnitude eigenvector of $\mathcal{B} = HDH$. Per tree we record the imbalance of the returned split, $\eta = \max(n_1, n_2)/\min(n_1, n_2)$, and whether that split is an edge of the true tree.

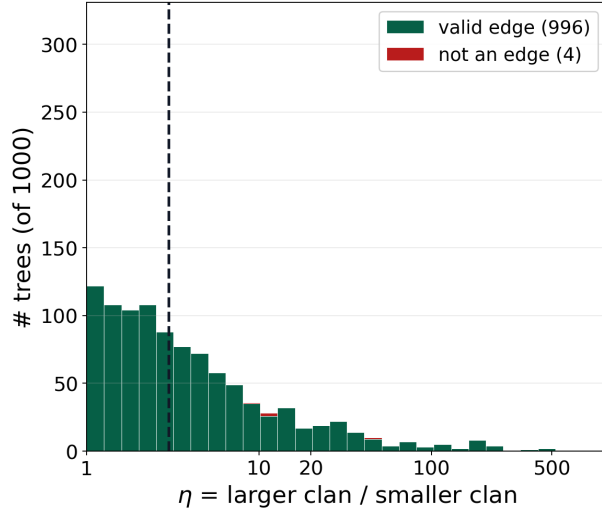
Table 2: Both operators on both cohorts, 1000 trees each.

	Kingman		birth–death	
	$L(S)$	$\mathcal{B}$	$L(S)$	$\mathcal{B}$
valid single-edge split	996	986	996	555
median η	3.00	1.85	3.08	1.53
maximum η	499	24.6	999	5.3
only $L(S)$ valid		14		443
only $\mathcal{B}$ valid		4		2
identical η		64.1%		33.2%

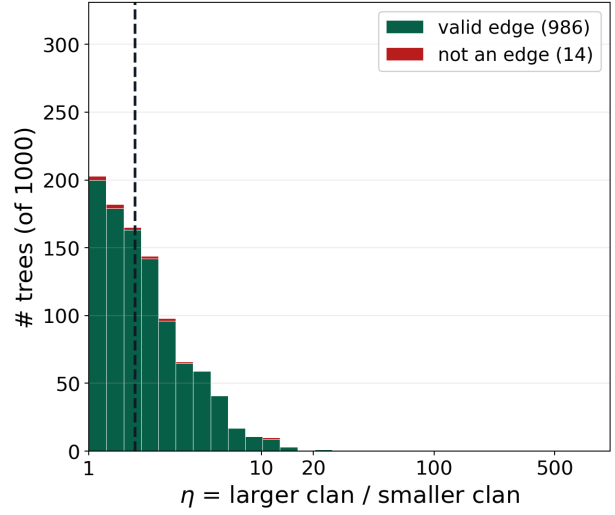
Four things follow.

1. **On Kingman trees the operators differ in *which* edge, not in whether they find one** — 99.6% and 98.6% valid, and the same split on 64.1% of trees.
2. **$\mathcal{B}$ ’s output is confined to near-balanced splits:** $\eta \leq 24.6$ on Kingman and $\eta \leq 5.3$ on birth–death, against true splits reaching $\eta = 999$. This is ?? — each edge contributes at least $2\tau_e n_1(e)n_2(e)/m$ to $|\lambda_1(\mathcal{B})|$, and $n_1(e)n_2(e)$ is maximal at balance. Where the rules disagree,

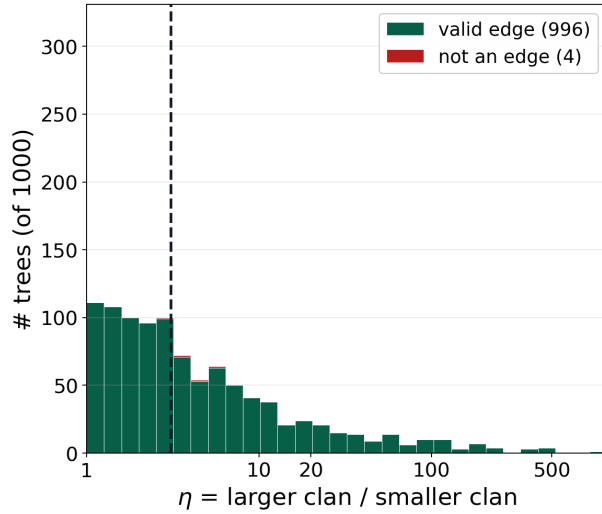
(a) Kingman, $L(S)$, k -means



(b) Kingman, $\mathcal{B}$, sign



(c) birth-death, $L(S)$, k -means



(d) birth-death, $\mathcal{B}$, sign

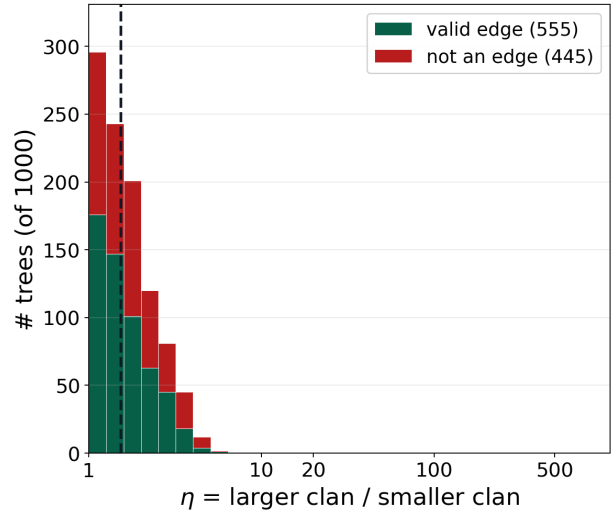


Figure 8: **Imbalance of the returned split, 1000 trees per model.** Green is a genuine single-edge bipartition of the true tree, red is not; dashed line is the median. All four panels share bins and axes.

$\mathcal{B}$'s split is the more balanced one in 355 of 359 Kingman cases and 654 of 668 birth–death cases.

3. **That confinement breaks $\mathcal{B}$ on imbalanced trees.** Its validity falls from 98.6% to 55.5% while $L(S)$ holds at 99.6%; on 443 birth–death trees $L(S)$ returns an edge and $\mathcal{B}$ does not. Lacking a near-balanced edge it does not decline — it returns a near-balanced *non*-edge.
4. **A narrow η distribution is a property of the operator, not of the tree.** Comparing the two routes' sampling rates is meaningful only with the target split held fixed; otherwise the rates describe different recovery problems, and on birth–death trees one is conditioned on a split absent from the tree.